

Grow360: Smart Employee Performance Tracker

L2 Assessment Development Plan



DAY 1: FOUNDATION & CORE FEATURES

Phase 1: Project Setup & Planning (4-5 hours)

Step 1.1: Project Architecture Design (90 minutes)

AI Prompts to Use:

Cursor/GitHub Copilot Prompt:

"Design a modern system architecture for an Employee Performance Management system called Grow360.

Requirements:

- Tech Stack: React frontend, Node.js/Express backend, PostgreSQL database
- Features: Employee profiles, Goal setting, Performance reviews, Skill assessment, Analytics
- Users: HR Admins, Managers, Employees
- Include: API layer, authentication, database design, deployment architecture
- Generate architecture diagram in Mermaid format"

AWS Q Developer Prompt:

"Review this architecture for security best practices:

- JWT authentication
- Role-based access control
- Database security
- API security measures

Suggest improvements and security hardening steps."

Expected Deliverables:

- ☐ System architecture diagram (Mermaid)
- ☐ Technology stack justification document
- ☐ Security architecture overview

Step 1.2: Database Schema with AI Assistance (90 minutes)

AI Prompts:

Cursor Prompt:

"Create a comprehensive PostgreSQL database schema for Grow360 employee performance system:

- Tables: employees, departments, roles, goals, performance_reviews, skills, teams
- Include proper relationships, constraints, indexes
- Add sample data inserts for a software company
- Include audit tables and user authentication tables
- Generate with proper foreign keys and data types"

Expected Deliverables:

- ☐ Complete database schema SQL file
- ☐ ERD diagram
- ☐ Sample data insertion scripts
- ☐ Database documentation

Step 1.3: API Endpoints Planning (60 minutes)

AI Prompt:

"Design RESTful API endpoints for Grow360 system:

- Authentication endpoints
- Employee management CRUD
- Goal management with progress tracking
- Performance review workflows
- Skills assessment APIs
- Analytics and reporting endpoints
- Include HTTP methods, request/response schemas, authentication requirements"

Expected Deliverables:

- ☐ API specification document
- ☐ Endpoint mapping with HTTP methods
- ☐ Request/response schemas

Step 1.4: Environment Setup (60 minutes)

AI Prompts:

Cursor Prompt:

"Set up a modern full-stack project structure for Grow360:

- Frontend: React with Vite, Tailwind CSS, React Query
- Backend: Node.js, Express, PostgreSQL, JWT authentication
- Include package.json, folder structure, environment configuration
- Add Docker setup for development
- Include ESLint, Prettier configuration"

Manual Steps:

1. Create GitHub repository: `grow360-performance-tracker`
2. Initialize frontend project with Vite
3. Initialize backend project with Express
4. Set up PostgreSQL database locally
5. Configure environment variables

Expected Deliverables:

- ☐ Project repository with proper structure
 - ☐ Development environment setup guide
 - ☐ Docker configuration files
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Phase 2: Backend Development (6-7 hours)

Step 2.1: Database Models Implementation (2 hours)

AI Prompts:

Cursor Prompt:

"Create Sequelize/Prisma models for Grow360 database:

- Employee model with relationships to Department, Role, Manager
- Goal model with categories, milestones, progress tracking
- PerformanceReview model with 360-degree feedback support
- Skill model with employee skill assessments
- Include model validations, associations, and methods
- Add proper TypeScript interfaces"

Tasks:

- ☐ Install and configure ORM (Sequelize/Prisma)
- ☐ Create all database models

- ☐ Set up model relationships
- ☐ Create database migrations
- ☐ Add model validations

Step 2.2: Authentication & Authorization (2 hours)

AI Prompts:

GitHub Copilot Prompt:

"Implement JWT-based authentication system:

- User registration and login endpoints
- Password hashing with bcrypt
- JWT token generation and validation
- Role-based middleware (HR Admin, Manager, Employee)
- Session management and logout
- Include TypeScript interfaces and error handling"

Tasks:

- ☐ Implement user authentication endpoints
- ☐ Create JWT middleware
- ☐ Implement role-based access control
- ☐ Add password hashing
- ☐ Create authentication tests

Step 2.3: Core API Development (3 hours)

AI Prompts:

Cursor Prompt for Each Module:

"Create Express.js API endpoints for Employee Management:

- GET /api/employees - List employees with pagination, filtering
- POST /api/employees - Create new employee
- GET /api/employees/:id - Get employee details
- PUT /api/employees/:id - Update employee
- DELETE /api/employees/:id - Soft delete employee
- Include validation, error handling, authorization middleware
- Add proper TypeScript types and JSDoc comments"

Repeat similar prompts for:

- Goals management APIs
- Performance review APIs
- Skills assessment APIs
- Analytics APIs

Tasks:

- ☐ Employee management endpoints
- ☐ Goal management endpoints
- ☐ Performance review endpoints
- ☐ Skills assessment endpoints
- ☐ Analytics endpoints
- ☐ Input validation middleware
- ☐ Error handling middleware

Step 2.4: Testing & Validation (1 hour)

AI Prompts:

Cursor Prompt:

"Create comprehensive API tests using Jest and Supertest:

- Authentication flow tests
- CRUD operation tests for all endpoints
- Authorization tests (role-based access)
- Input validation tests
- Error handling tests
- Include test data setup and teardown
- Add test coverage reporting"

Tasks:

- ☐ Set up testing framework
- ☐ Create API endpoint tests
- ☐ Test authentication flows
- ☐ Test authorization rules
- ☐ Generate test coverage reports

Expected Deliverables:

- ☐ Complete backend API
- ☐ API documentation (auto-generated)

- ☐ Test suite with coverage report
 - ☐ Postman collection for testing
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Phase 3: Frontend Foundation (4-5 hours)

Step 3.1: UI Framework Setup (90 minutes)

AI Prompts:

Cursor Prompt:

"Set up modern React application for Grow360:

- Vite + React + TypeScript setup
- Tailwind CSS for styling
- React Router for navigation
- React Query for API calls
- Context API for state management
- React Hook Form for forms
- Include folder structure: components, pages, hooks, services, types
- Add common UI components: Layout, Navigation, Forms"

Tasks:

- ☐ Initialize React project with TypeScript
- ☐ Configure Tailwind CSS
- ☐ Set up routing structure
- ☐ Configure React Query
- ☐ Create base layout components
- ☐ Set up state management

Step 3.2: Authentication UI (90 minutes)

AI Prompts:

Cursor Prompt:

"Create authentication components for Grow360:

- Login form with email/password validation
- Protected route wrapper component
- Authentication context with login/logout functions
- Role-based component rendering
- Loading states and error handling
- Responsive design with Tailwind CSS
- Include proper TypeScript interfaces"

Tasks:

- ☐ Login/logout components
- ☐ Protected route wrapper
- ☐ Authentication context
- ☐ Role-based navigation
- ☐ Loading and error states

Step 3.3: Core Components Development (2-3 hours)

AI Prompts:

Cursor Prompt for each module:

"Create React components for Employee Management:

- EmployeeList component with search, filter, pagination
- EmployeeForm for create/edit operations
- EmployeeProfile showing detailed information
- Use React Hook Form for form handling
- Include proper validation and error messages
- Responsive design with Tailwind CSS
- TypeScript interfaces for all props and data"

Repeat for:

- Goal management components
- Performance review components
- Skills assessment components
- Dashboard components

Tasks:

- ☐ Employee management components

- ☐ Goal management components
- ☐ Performance review components
- ☐ Skills assessment components
- ☐ Dashboard components
- ☐ Common UI components (tables, forms, modals)

Step 3.4: API Integration (60 minutes)

AI Prompts:

Cursor Prompt:

"Create API service layer for Grow360 frontend:

- API client with axios configuration
- Authentication headers handling
- Error handling and retry logic
- Type-safe API functions for all endpoints
- React Query hooks for data fetching
- Loading states and error boundaries
- Include proper TypeScript interfaces"

Tasks:

- ☐ API client configuration
- ☐ API service functions
- ☐ React Query hooks
- ☐ Error handling
- ☐ Loading states management

Expected Deliverables:

- ☐ Functional frontend application
- ☐ Authentication flow working
- ☐ Core CRUD operations implemented
- ☐ Responsive design
- ☐ Error handling and loading states



DAY 2: ADVANCED FEATURES & POLISH

Phase 4: Advanced Features (4-5 hours)

Step 4.1: Business Logic Implementation (2 hours)

AI Prompts:

Cursor Prompt:

"Implement advanced business logic for Grow360:

- Goal progress calculation and status updates
- Performance review workflow (draft → submitted → reviewed → completed)
- Skill assessment scoring and recommendations
- Notification system for due dates and reminders
- Reporting and analytics calculations
- Include proper error handling and validation"

Tasks:

- ☐ Goal progress tracking logic
- ☐ Review workflow implementation
- ☐ Skill assessment algorithms
- ☐ Notification system
- ☐ Business rule validations

Step 4.2: Dashboard & Analytics (2-3 hours)

AI Prompts:

Cursor Prompt:

"Create comprehensive analytics dashboard:

- Goal completion rate charts using Chart.js/Recharts
- Performance trend visualizations
- Team performance comparisons
- Skill gap analysis charts
- KPI cards with key metrics
- Responsive design with interactive charts
- Export functionality for reports"

Tasks:

- ☐ Analytics dashboard page
- ☐ Chart components (goals, performance, skills)
- ☐ KPI metrics calculation
- ☐ Data visualization components

Phase 5: Security & Performance (2-3 hours)

Step 5.1: Security Implementation (90 minutes)

AWS Q Developer Prompts:

"Review and enhance security for Grow360 application:

- Input validation and sanitization
- SQL injection prevention
- XSS protection
- CSRF protection
- Rate limiting implementation
- Security headers configuration
- Audit logging for sensitive operations"

Tasks:

- ☐ Input validation enhancement
- ☐ Security middleware implementation
- ☐ Rate limiting
- ☐ Security headers
- ☐ Audit logging
- ☐ Security testing

Step 5.2: Performance Optimization (90 minutes)

AI Prompts:

Cursor Prompt:

"Optimize Grow360 application performance:

- Database query optimization with indexes
- React component optimization (memo, useMemo, useCallback)
- Image optimization and lazy loading
- Bundle optimization and code splitting
- Caching strategies (Redis for backend, React Query for frontend)
- Performance monitoring setup"

Tasks:

- ☐ Database query optimization
 - ☐ React performance optimization
 - ☐ Bundle optimization
 - ☐ Caching implementation
 - ☐ Performance monitoring
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Phase 6: Deployment & Documentation (2-3 hours)

Step 6.1: Deployment Setup (90 minutes)

AI Prompts:

Cursor Prompt:

"Set up deployment for Grow360:

- Frontend: Deploy to Vercel with environment variables
- Backend: Deploy to Railway/Render with PostgreSQL
- Database: Set up production PostgreSQL instance
- Environment configuration for production
- CI/CD pipeline with GitHub Actions
- Domain configuration and SSL setup"

Tasks:

- ☐ Production environment setup
- ☐ Frontend deployment (Vercel)
- ☐ Backend deployment (Railway/Render)
- ☐ Database setup (PostgreSQL)
- ☐ Environment variables configuration
- ☐ CI/CD pipeline

Step 6.2: Final Documentation (90 minutes)

AI Prompts:

Cursor Prompt:

"Create comprehensive documentation for Grow360:

- User guide with screenshots for each feature
- Installation and setup instructions
- API documentation with examples
- Deployment guide
- Troubleshooting section
- Architecture overview
- Contributing guidelines"

Tasks:

- ☐ User guide with screenshots
 - ☐ Technical documentation
 - ☐ API documentation
 - ☐ Deployment guide
 - ☐ README file update
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REQUIRED SUBMISSIONS

1. Application Deliverables

Live Application URL

- ☐ Deploy frontend to Vercel
- ☐ Deploy backend to Railway/Render
- ☐ Ensure all features are working
- ☐ Test with sample data

GitHub Repository

- ☐ Create comprehensive README.md
- ☐ Include installation instructions
- ☐ Add screenshots and demo GIFs
- ☐ Clean up code and add comments
- ☐ Include .env.example files

Demo Video (5 minutes)

- ☐ Screen recording of key features
- ☐ Show login as different user types
- ☐ Demonstrate CRUD operations

- ☐ Show analytics dashboard
- ☐ Upload to YouTube/Loom

Admin Credentials

- ☐ Create test accounts for evaluation
- ☐ Document credentials in README
- ☐ Include different role types

2. Development Process Documentation

Create: `docs/development-process.md`

markdown

Development Process Report

Project Overview

- **Project Chosen**: Employee Performance Management System (Grow360)
- **Technology Stack**:
 - Frontend: React, TypeScript, Tailwind CSS, Vite
 - Backend: Node.js, Express, PostgreSQL, JWT
 - Deployment: Vercel (Frontend), Railway (Backend)
- **Development Timeline**:
 - Day 1: 10 hours (Setup + Backend + Frontend Foundation)
 - Day 2: 8 hours (Advanced Features + Deployment)

AI Tool Usage Summary

- **Cursor**: 9/10 - Excellent for component generation and business logic
- **GitHub Copilot**: 8/10 - Great for API endpoints and utility functions
- **AWS Q Developer**: 7/10 - Helpful for security recommendations

Architecture Decisions

- **Database Design**: PostgreSQL with proper relationships and indexes
- **API Architecture**: RESTful APIs with JWT authentication
- **Frontend Architecture**: Component-based with React Query for state management

Challenges & Solutions

- **Technical Challenges**: Complex role-based access control
- **AI Limitations**: Required manual fine-tuning for business logic
- **Breakthrough Moments**: AI-generated database schema saved 4+ hours

3. AI Prompt Library

Create: `docs/ai-prompt-library.md`

Document all prompts used with:

- Context provided
- Output quality rating (1-10)
- Iterations needed
- Final implementation

4. Reflection Report

Create: `docs/reflection-report.md`

- Key learnings from AI development
 - Most effective AI techniques
 - Challenges faced and solutions
 - Future application strategies
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SUCCESS CHECKLIST

Technical Implementation (50 points)

- ☐ All core features working (Employee, Goals, Reviews, Skills)
- ☐ Clean, maintainable code with proper structure
- ☐ Effective AI tool integration throughout development
- ☐ Proper error handling and validation

Process Mastery (30 points)

- ☐ Followed structured development approach
- ☐ Used AI effectively for problem-solving
- ☐ Completed within 2-day timeline
- ☐ Proper version control and commits

Documentation & Knowledge Transfer (20 points)

- ☐ Comprehensive development process documentation
 - ☐ Complete AI prompt library with ratings
 - ☐ Thoughtful reflection on AI effectiveness
 - ☐ Clear setup and deployment instructions
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TIME ALLOCATION GUIDE

Day 1 (10 hours):

- Phase 1: Project Setup (4-5 hours)
- Phase 2: Backend Development (6-7 hours)
- Phase 3: Frontend Foundation (4-5 hours)

Day 2 (8 hours):

- Phase 4: Advanced Features (4-5 hours)
- Phase 5: Security & Performance (2-3 hours)
- Phase 6: Deployment & Documentation (2-3 hours)

Total: 18 hours over 2 days



GETTING STARTED

1. **Set up your development environment**
2. **Start with the architecture design using AI**
3. **Follow each phase sequentially**
4. **Document everything as you go**
5. **Test frequently and iterate**
6. **Deploy early and often**

Good luck with your L2 Assessment! 🎉